WORKSHEET

Chapter 3: One-to-one functions

1 For each of the following, determine whether the function is one-to-one or not one-to-one.

a $f(x) = 7 - 5x$

b $f(x) = 2^x$

c $f(x) = x^3 - x$

d $f(x) = \dfrac{1}{5 - x}$

e $f(x) = e^x + 5$

f $f(x) = x^2 - x^3 + 10x - 5$

2 Which of the following functions is one-to-one?

a $f(x) = 3x^2 - x - 8$

b $f(x) = \dfrac{1}{7x + 5}$

c $f(x) = |2x - 5|$

d $f(x) = 2x - 3\sqrt{2}$

e $f(x) = \sqrt{x - 1}$

f $f(x) = x^3 + 4$

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3 For each of the following functions, show it is increasing, decreasing or neither, and hence determine if it is one-to-one.

a $f(x) = 3x^2 + 5x - 12$

b $f(x) = x^3 + \frac{1}{2}x$

c $f(x) = \dfrac{4}{x + 1}$

d $f(x) = -x^5 - x^3 - 2x$

e $f(x) = |4x + 9|$

f $f(x) = -\sqrt{x + 4}$

4 Is the function $f(x) = \dfrac{x - 3}{x + 2}$ one-on-one?

5 Show that the function $f(x) = x^3 - 12x$ on the interval $[-2, 2]$ is one-on-one.

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6 Is the function $f(x) = 1 - x^{\frac{1}{3}}$ one-on-one?

7 Is the function $f(x) = x^3 + x^2 - 6x$ on the interval $[-3, 2]$ one-on-one?

8 Is the function $f(x) = (x - 1)^3 + x^2$ one-on-one?

9 Is the function $f(x) = (4 - x)^3 + 6$ one-on-one?

10 Is the function $f(x) = \frac{x^3}{3} - \frac{x^2}{2} - 12x + 4$ on the interval $[-3, 4]$ one-on-one?

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Answers

1 **a** One-on-one

b One-on-one

c Not one-on-one

d One-on-one

e One-on-one

f Not one-on-one

2 **b**, **d**, **e** and **f** are one-on-one functions.

3 **a** Neither increasing nor decreasing, not one-to-one

b Increasing, one-to-one

c Decreasing, one-to-one

d Decreasing, one-to-one

e Neither increasing nor decreasing, not one-to-one

f Decreasing, one-to-one

4 Yes (increasing function)

5 The function is always decreasing over the interval, so it is one-on-one.

6 Yes (decreasing function)

7 No (neither increasing nor decreasing)

8 Yes (increasing function)

9 Yes (decreasing function)

10 Yes (decreasing function)

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